→ Harmonic oscillator → 2 (i) (a) → Prob part

PYL122 Assignment 6
Tobias Toll

Instructions: *Solve the problems during the tutorial session. Hand in the solutions to the TA one day before the following tutorial. Submit hardcopy to MS207c-19, or soft copy to gaurab.Kumar.Dash@physics.iitd.ac.in. Assignment will be considered for marks if handed in on time.*

Nice. Done Jeff.

1. (a) Show that the operators $\hat{a}_{\pm} = \hat{p} \pm im\omega\hat{x}$, when operating on a solution to the Schrödinger equation for the harmonic oscillator potential $V(x) = \frac{1}{2}kx^2$, steps the energy of the solution up or down by an energy quantum of $\hbar\omega$.
(b) Find the lowest energy eigenvalue E_0, and its wavefunction ϕ_0.
(c) Show that the (unnormalised) state with energy $E_1 = E_0 + \hbar\omega$ can be written $\phi_1 = 2i\hbar\alpha x\phi_0$, with $\alpha = \frac{m\omega}{\hbar}$. Show, by operating with the Hamiltonian, that this function has the expected energy eigenvalue.
(d) Find ϕ_2 using the same method.
***(e)** Use the computer program of Assignment 5 for the harmonic oscillator and show that your resulting wave function approximates the true solution.

2. For a classical harmonic oscillator, the energy is given by

$$E_{\text{classical}} = \frac{1}{2}mv^2 + \frac{1}{2}m\omega^2x^2$$

At the turning point, where x is at its maximum value, the velocity in the classical potential is zero. Solving for this we get:

$$x_{\max} = \sqrt{\frac{2E}{m\omega^2}}$$

(a) Plug in the value for the quantum mechanical energy E_n, and plot the solutions ϕ_0, ϕ_1, and ϕ_2, as well as their probabilities $|\phi_0|^2$, $|\phi_1|^2$, and $|\phi_2|^2$. Estimate the probability for $|x| > |x_{\max}|$ (order of magnitude). Note that this probability, which is zero in the classical case, is significant in quantum mechanics.
(b) Suppose that we have a classical harmonic oscillator, oscillating back and forth. The probability of finding it in an interval Δx is given by $\Delta t/T$, where the period $T = 2\pi/\omega$, and Δt is the amount of time the particle spends in the interval Δx. In each period, the particle will pass through Δx twice. Therefore, $\Delta t = 2\Delta x/v$, where $v = x_{\max}\omega\cos(\omega t)$ is the velocity of the particle.
Show that the "Classical Probability Density" can be written:

$$|\psi(x)|^2_{\text{classical}} = \frac{1}{\pi\sqrt{x_{\max}^2 - x^2}}$$

→ Easy

(c) Plot this curve in the same diagrams as the probability plots in **(a)**. How do you expect the two curves to compare as $n \to \infty$? See Figure 2.7 in Griffiths as a reference.
(d) For a classical pendulum, the energy is $E = mgh$, where h is the maximum height of the weight of mass m. What are the differences in height for adjacent quantum energy levels for a typical weight attached to a string? Is it something which is measurable?

Easy

Practice problems (not included in assignment):
Griffiths 2.11, 2.12, 2.13